TEACHER GUIDE · GRADE 1 · INTRO DEBUGGING

What Is a Bug?

Quick facilitation notes & answer key

Goal

Students learn that a bug is code that produces the wrong result, and that you find one by running the program and comparing the result to the target. They run a correct program and a buggy one and decide which has the bug. Reading code and judging whether its outcome is right is the first step of C3.2 (and of debugging).

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 with a marker on 4 (the target) so students can walk each program.

Run it (about 20 min)

1. I DO: run Program A — 'start 1, forward 3 lands on 4 — that is the target, so no bug.'
2. WE DO: Exercise 1 — A reaches 4 (no bug). Exercise 2 — run Program B: start 1, forward 1 lands on 2, not 4 — there is a bug.
3. YOU DO: Exercise 3 — Program B has the bug; a bug is code that does the wrong thing.
4. Connect: 'Coders run their code and check the result — that is how they spot bugs.'

Answer key (modelled by the run-gate)

Ex 1 — Program A: 1 → forward 3 → 4. It reaches the target, so there is NO bug.

Ex 2 — Program B: 1 → forward 1 → 2. It does NOT reach 4 (it lands on 2), so there IS a bug.

Ex 3 — Program B has the bug. A bug is a mistake in the code that makes it do the wrong thing (here, Bit lands on 2 instead of 4).

Watch for & differentiate

Common idea: students may say B is 'broken' — affirm and give the word 'bug' for a code mistake.

Simplify: walk both programs on the floor path so the miss in B is felt.

Extend: ask how far off Program B is from the target (2 short) — this previews the missing-block fix.

Success indicator: the child runs both programs, says A reaches 4 with no bug and B lands on 2 with a bug, and explains what a bug is.